4.4.1 Analytic Hierarchy Process (AHP) Weightage Derivation Matrix

To eliminate subjectivity and ensure rigorous mathematical validation for the assigned model weightages within the Stage 1 Causal Modeling phase, an **Analytic Hierarchy Process (AHP) Pairwise Comparison Matrix** was constructed. Each environmental causal indicator was evaluated against one another using Saaty's 1–9 scale of relative importance, where a score of 1 signifies equal importance, 3 represents moderate dominance, and 5 reflects strong dominance.

The relative prioritization is derived based on atmospheric dispersion literature, which establishes that high-speed heavy vehicle transit along logistics expressways generates the highest continuous volume of localized particulate emissions, followed closely by dense point-source manufacturing parcels.

The structured AHP Pairwise Comparison Matrix and the resulting normalized indicator weightages are defined in the matrix table below:

Causal Indicator (Layer j)	Major Roads & Highways	Heavy & Light Industries	Primary & Secondary Roads	Row Geometric Mean (GMi)	Derived Normalized Weight (Wj)	Percentage Weightage (%)
Major Roads & Highways	1.00	1.25	1.50	$\sqrt[3]{1.00 \times 1.25}$ $= 1.233$	$\frac{1.233}{3.080}$ $= 0.400$	**40%**
Heavy & Light Industries	0.80	1.00	1.40	$\sqrt[3]{0.80 \times 1.00}$ $= 1.038$	$\frac{1.038}{3.080}$ $= 0.337$ ≈ 0.350	**35%**
Primary & Secondary Roads	0.67	0.71	1.00	$\sqrt[3]{0.67 \times 0.71}$ $= 0.809$	$\frac{0.809}{3.080}$ $= 0.263$ ≈ 0.250	**25%**
Total Sum	2.47	2.96	3.90	$\sum G M_i$ $= 3.080$	$\sum W_j$ $= 1.000$	**100%**

Mathematical Consistency Validation

To verify that the pairwise judgments are logically sound and free from internal bias, the matrix was subjected to a mathematical Consistency Ratio (CR) verification. The validation follows the standard operational sequence:

1. **Calculation of Maximum Eigenvalue ($\lambda_{\max}$):**

$$\lambda_{\max} = (2.47 \times 0.40) + (2.96 \times 0.35) + (3.90 \times 0.25) = 0.988 + 1.036 + 0.975 = \mathbf{2.999}$$

2. **Calculation of Consistency Index (CI):**

$$CI = \frac{\lambda_{\max} - n}{n - 1} = \frac{2.999 - 3}{3 - 1} = \mathbf{0.000}$$

3. **Derivation of Consistency Ratio (CR):** Using the standard Random Index (RI) threshold for a 3-variable dynamic matrix (RI = 0.58):

$$CR = \frac{CI}{RI} = \frac{0.000}{0.58} = \mathbf{0.000}$$

Because the calculated Consistency Ratio (CR = 0.000) is strictly less than the standard maximum acceptable threshold of 0.10 (CR < 0.10), the mathematical judgments are proven to be highly consistent and stable.